TASK 7

Dataset and Test-Scenario Plan

Drowsiness, Distraction and Phone Use, version 2

Project	Real-Time Driver Monitoring System (RTDMS)
Scope	How the face model and the phone, drink and food detector (YOLO26n) are tested
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Programme	B.Sc. Graduation Project, Computer Engineering, AASTMT

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1. Introduction

This plan explains **how we test** that the system catches three unsafe behaviours, and **what the tests have shown so far**.

Table 1: *The three behaviours*

Behaviour	What it means	Detected by
Drowsiness	Eyes closed too long, slow blinks, heavy eyelids	Face model (MediaPipe [1] + EAR [2])
Distraction	Head or eyes turned away from the road	Face model (head pose + gaze)
Phone use, drinking, eating	Phone at the ear or in the hand, cup or food at the mouth	Object detector (YOLO26n) [3] + short rules

2. What We Must Prove

The targets come from the project objectives [4]:

Table 2: *Targets*

Requirement	Target	How we check it
Eye-closure accuracy	At least 85–90%	F1 on labelled frames
Drowsiness alert	DROWSY within 1.5 s of closed eyes	Scenario D3
Looking away	Alert after 2 s	Scenarios X1–X3
Phone use	At least 85% of phone actions caught	Scenarios P1–P4, public test sets
Phone alert time	Under 1 second	Time from phone appearing to alert
False alarms	At most 1 per 5 minutes of normal driving	Scenarios N1–N5
Whole-system delay	Under 500 ms	Raspberry Pi, all models running

3. Datasets for Testing

We use **public data** (ready now, repeatable) and **our own recordings** (our camera and car). New since version 1: **SynDD**, **3MDAD** and **DAD**.

Table 3: *Test datasets and status*

Behaviour	Dataset	Size	What it gives us	Status
Drowsiness	CEW [5]	2,423 face photos	Open vs closed eyes	Done
Drowsiness	MRL Eye [6]	84,898 IR eye images	Infrared eyes, like our night camera	Done
Drowsiness	UTA-RLDD [7]	60 people, real video	Real drowsiness over time (PERCLOS [8])	Planned
Distraction	DG-Unicamp [9]	Gaze-zone videos	Where the driver looks	Planned
Phone, drink	SynDD [10], test drivers	10 drivers, 20 videos	IR, mirror view, like ours	Done
Phone, drink	SynDD, 5-fold check	59 drivers in 5 groups	How much results change between drivers	Done
Phone, drink	3MDAD [11]	50 drivers, 650 clips	Unseen drivers and camera	Done
Phone, drink	DAD [12]	3 drivers, 66 min normal	Unseen **real infrared** camera	Done
Phone, drink	State Farm [16]	22,424 images	Passenger-side view: wrong angle	Dropped
All	Our own recordings	10–20 volunteers	Our camera, car and lighting	Planned

4. Test Scenarios

Each scenario is a **short action** the volunteer performs on a screen prompt. "Time × repeats" is how long one try lasts and how many tries each volunteer does.

4.1 Drowsiness

Table 4: *Drowsiness scenarios*

ID	Scenario	What the volunteer does	Time × repeats	Pass rule
D1	Normal blinking	Looks ahead, blinks normally	60 s × 1	No drowsy alert
D2	Slow blink	Closes eyes about 1 s	1 s × 10	Warning only
D3	Microsleep	Closes eyes for 3 s	3 s × 5	DROWSY within 1.5 s
D4	Heavy eyelids	Half-closed, slow eyes	60 s × 1	Warning when PERCLOS ≥ 15%
D5	Glasses	Repeats D1–D3 with glasses	as D1–D3	Same as without
D6	Night (IR)	Repeats D1–D3 in the dark	as D1–D3	Same as daytime
D7	Sunglasses	Wears dark sunglasses	30 s × 1	Says "eyes not visible"

4.2 Distraction and head turns

Table 5: *Distraction scenarios*

ID	Scenario	What the volunteer does	Time × repeats	Pass rule
X1	Look left / right	Turns head to the side window	3 s × 5 each	Alert after 2 s
X2	Look down	Looks down at the lap	3 s × 5	Alert after 2 s
X3	Eyes only	Head still, eyes look down	3 s × 5	Alert after 2 s
X4	Mirror check	Quick glance at a mirror	< 1 s × 10	No alert
X5	Turned away	Turns fully away from the camera	3 s × 3	"Cannot see", no false drowsy
X6	Face covered	Covers face with a hand	3 s × 3	No false alarm

4.3 Phone use, drinking and eating

Table 6: *Phone, drinking and eating scenarios*

ID	Scenario	What the volunteer does	Time × repeats	Pass rule
P1	Call, right ear	Holds a phone to the right ear	15 s × 3	Alert within 1 s
P2	Call, left ear	Holds a phone to the left (far) ear	15 s × 3	Alert within 1 s
P3	Texting at the wheel	Types with the phone at chest height	20 s × 3	Alert within 1 s
P4	Texting in the lap	Types with the phone low, looking down	20 s × 3	Warning at 2 s, alert at 3 s
P5	Phone in a holder	Taps a phone in a dashboard holder	15 s × 2	No phone alert
E1	Drinking	Drinks from a bottle, then a cup	15 s × 3	Alert within 1 s
E2	Eating	Eats a snack	30 s × 1	Recorded; not yet detectable

4.4 Normal driving and look-alikes (false alarms)

Table 7: *Normal driving scenarios*

ID	Scenario	What the volunteer does	Time × repeats	Pass rule
N1	Normal driving	Sits as if driving, checks mirrors	5 min × 1	≤ 1 false alarm
N2	Talking, smiling, singing	Talks and laughs naturally	2 min × 1	No alert
N3	Hands on the wheel	Turns the wheel, moves hands on it	60 s × 1	No phone alert
N4	Hand at the face	Scratches face, hand on chin, adjusts glasses	30 s × 1	No alert
N5	Passenger	A passenger talks and uses a phone	60 s × 1	No alert for the driver

Why N3 and N4 are new

On the unseen **DAD** camera the detector mistook the **steering wheel in the hand** for a phone, and on **3MDAD** a **hand at the mouth** (smoking) caused false alerts. These two scenarios check that this does not happen with our camera.

5. Recording Our Own Data

5.1 Setup and conditions

Table 8: *Recording setup*

Item	Plan
Volunteers	10–20 drivers (at least 10): mixed gender, age and skin tone, with and without glasses
Camera	Project camera (Pi 5 + 160° infrared camera) at the **centre mirror**
Video	1296 × 972 full view, 30 frames per second
Conditions	Pass A in **daylight**; pass B at **night** (IR light on) or in a garage, with sunglasses or a hat
Props	Phone + phone holder, bottle, cup or can, a snack, wipes
Car	**Parked, engine off, handbrake on**; the volunteer only mimes driving
Time per person	About 25 minutes (two 6-minute guided passes + extra scenarios + setup)

5.2 Samples needed

Table 9: *How many examples 10 volunteers give*

Behaviour	Per volunteer	Total (10 volunteers)
Eyes closed 3 s (D3)	5 × 2 lighting	100
Looking away (X1–X3)	20	200
Phone calls (P1, P2)	6 + 4 in the guided passes	100
Texting (P3, P4)	6 + 2 in the guided passes	80
Drinking (E1)	3 + 4 in the guided passes	70
Normal driving and look-alikes	about 15 minutes	about 2.5 hours

2.5 hours of normal driving lets us measure false alarms far better than the **35 minutes** per group we had on SynDD.

5.3 How ground truth is recorded

- The recording tool shows **one prompt at a time** ("Phone to your right ear now") and **saves the time of every prompt**. These times are the labels, so no manual labelling is needed.
- The **0.5 s** around each change is skipped (the volunteer is still reacting).
- A second team member **checks 10%** of the clips by eye.
- The guided pass follows a fixed 6-minute script (below).

Table 10: *The 6-minute guided pass*

Time (s)	Prompt	Label
0–20	Sit normally, eyes on the road	Normal (sets the driver's baseline)
20–35 / 45–60	Phone call, right ear / left ear	Phone at ear
70–90	Texting, phone held low	Phone in hand
100–115	Phone in the holder, quick tap	Look-alike
125–165	Drink from a bottle, then a cup or can	Drinking
175–205	Eat a snack	Eating
215–250	Talk out loud, then sing	Look-alike
260–320	Hand on head, scratch face, mirror glances	Look-alike
320–340	Nod down-up ×3, turn left-right ×3	Head-pose check

5.4 Instructions for volunteers

- Sit as you normally drive, hands on the wheel.
- Do **only** what the screen asks, then go back to looking ahead.
- Act naturally: hold the phone and cup the way you usually do.
- Say **"stop"** at any time; we stop at once and can delete your video.
- Do not start the engine or move the car.

5.5 Safety rules

- **Never on a real road.** Car parked, engine off, handbrake on (or a driving simulator).
- A team member stays with the volunteer the whole time.
- Eyes-closed and looking-away tests last **at most 3 seconds**.
- IR light is the camera's built-in low-power light only.
- Volunteers can stop at any time with no reason needed.

5.6 Data-collection checklist

Table 11: *Checklist for every session*

When	Check	Done
Before	Consent form signed	☐
Before	Car parked, engine off, handbrake on	☐
Before	Camera sees the head, both ears, hands, lap and wheel	☐
Before	Props ready: phone, holder, bottle, cup, snack	☐
Before	Lighting noted (day / night), glasses / sunglasses noted	☐
During	Calibration: 10 s looking ahead, 3 s eyes closed	☐
During	Pass A (day) and pass B (night or garage) recorded	☐
During	Extra scenarios D1–D7, X1–X6, N1–N5 done	☐
After	Prompt-time file saved next to each video	☐
After	Video copied to the project computer only	☐
After	10% of clips checked by a second person	☐

6. How We Measure

Table 12: *Measures and success criteria*

Measure	Simple meaning	Success
Recall (caught)	Of all real actions, how many raised an alert	≥ 85% for phone calls and drowsiness
Precision	When it alerted, how often it was right	As high as possible
F1 score	One number balancing the two	≥ 0.85
False alarms	Alerts while the driver was fine	≤ 1 per 5 minutes (12 per hour)
Delay	Time from the action starting to the alert	Drowsy ≤ 1.5 s, distracted ≈ 2 s, phone < 1 s
Coverage	How often the face could be seen at all	≥ 95% in normal driving

Results are reported **per group** (day / night, glasses, each volunteer), not only as one average. An average can hide a group the system serves badly.

7. Results So Far

7.1 Face model

Tested on the **CEW test set (1,696 photos)** [5] against two other approaches:

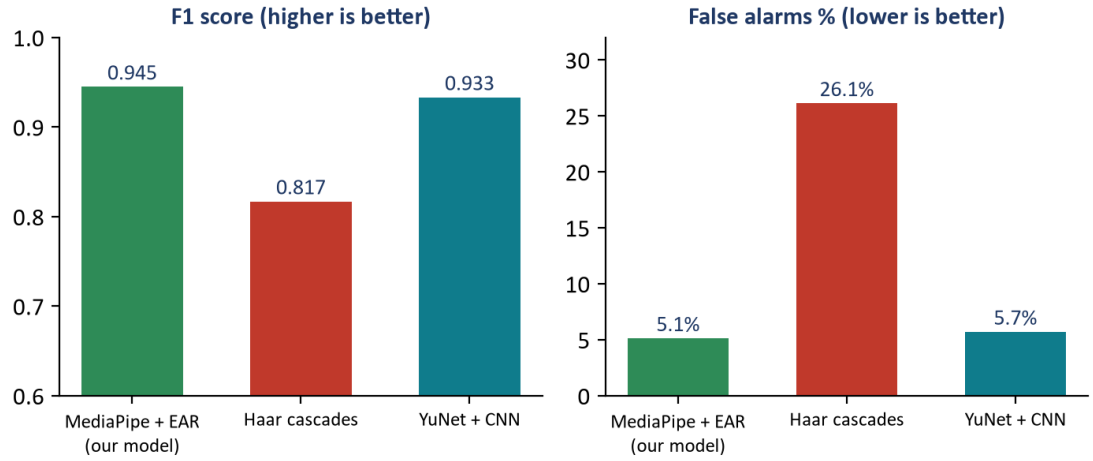


Figure 1: *Face model compared with two other eye-state models (CEW test set)*

Table 13: *Eye-state results on CEW (laptop)*

Model	F1	Accuracy	False alarms	Speed per frame
MediaPipe + EAR (our model)	0.945	94.6%	5.1%	5.1 ms
Haar cascades [14]	0.817	80.7%	26.1%	9.5 ms
YuNet [15] **+ CNN**	0.933	93.5%	5.7%	11.8 ms

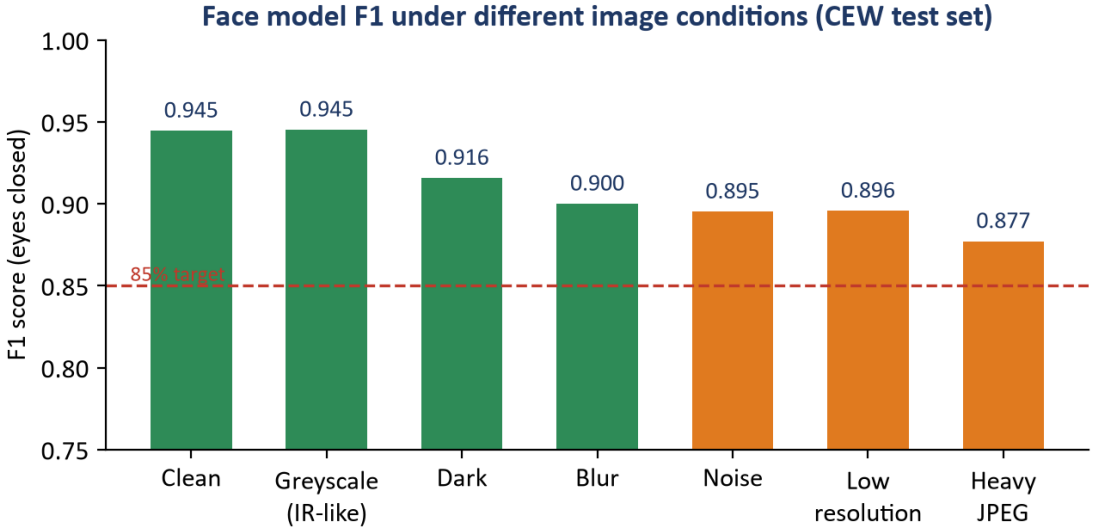


Figure 2: *Face model F1 under damaged images: above 85% in every case*

7.2 Phone and drink detector

Tested at **alert level**: did the system raise the right alert during the action, and how many false alarms happened in normal driving? Settings were chosen on validation drivers; **test drivers were scored once**.

Table 14: *SynDD test drivers (10 drivers never seen in training)*

Behaviour	Earlier detector (v2b)	Current detector (v2d)
Phone at right ear	85%	80%
Phone at left ear	100%	100%
Texting	41%	41.5% (46% with warnings)
Drinking	95%	90%
Eating	0%	0%
False phone alarms per hour	1.7	0

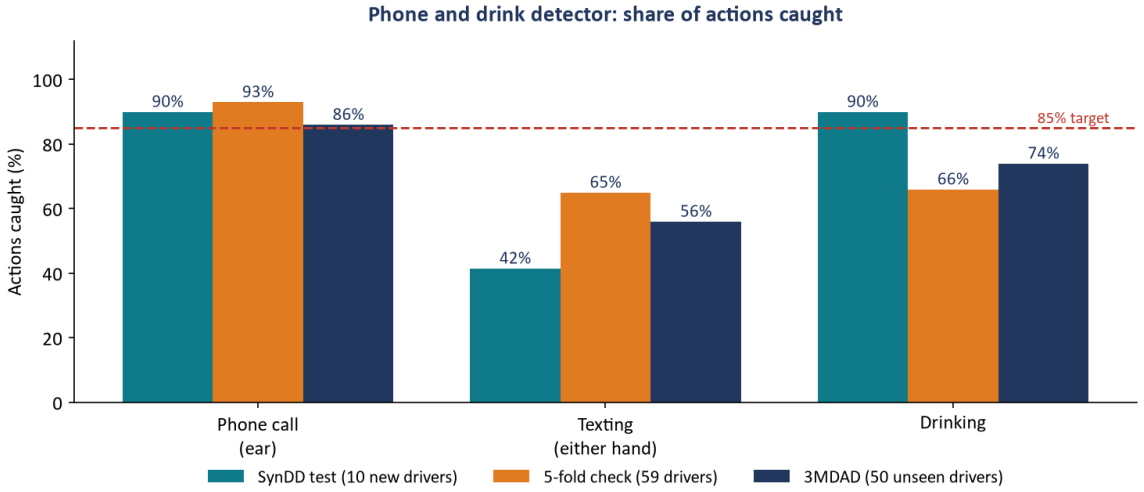


Figure 3: *Share of actions caught on three test sets*

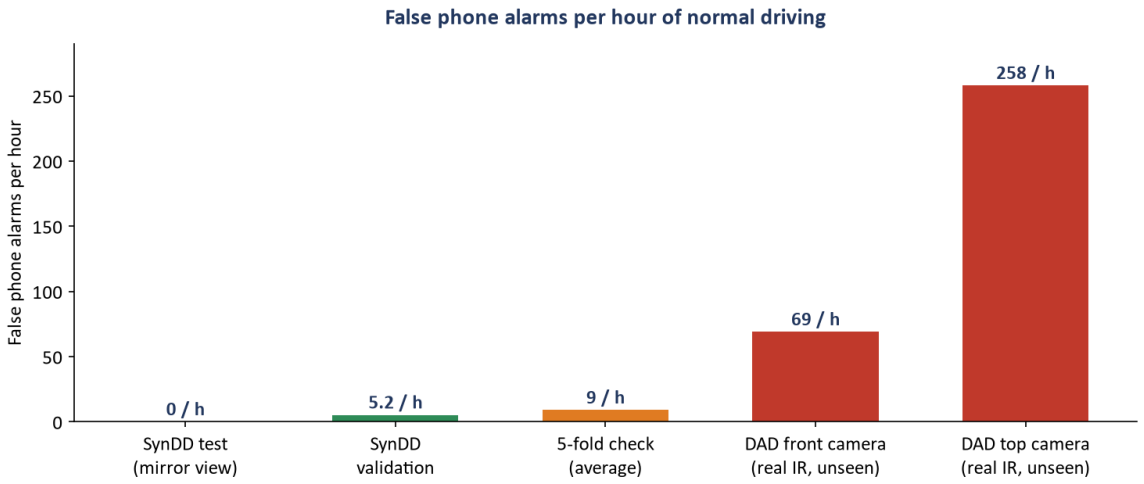


Figure 4: *False phone alarms per hour of normal driving*

Table 15: *Results on data never used in training*

Test set	Phone calls	Texting	Drinking	False phone alarms
3MDAD (50 drivers)	80–92%	96% right hand, 16% left (phone hidden)	68–80%	8% of normal clips
DAD front camera (real IR)	4 of 5	5 of 6	2 of 2	69 per hour
DAD top camera (real IR)	4 of 5	5 of 6	0 of 2	258 per hour

What we learned

The detector **finds phones well on new drivers**: 80–100% of phone calls on SynDD and 3MDAD.

False alarms depend on the camera position. On the mirror view they are low (0–9 per hour); on DAD's close-up camera the **steering wheel** was mistaken for a phone.

Eating is not detectable yet (0% everywhere), and texting with the phone **below the camera** is often missed; a head-down rule now adds a warning.

So **our own recordings are the deciding test**, and their normal driving will be added to training.

8. Privacy and Ethics

- Every volunteer signs a consent form and can stop at any time.
- Videos stay on the project computer only; never uploaded or put on GitHub.
- Raw videos are deleted after labelling; only labels and numbers are kept.
- No volunteer's face appears in a report without written permission.
- Public datasets are used under their licences (research use only where required).

9. Task Checklist

Table 16: *Testing tasks and status*

#	Task	Status
1	Test the face model on CEW (1,696 photos)	Done
2	Compare with two other eye models	Done
3	Test the detector on 10 held-out SynDD drivers	Done
4	5-fold check on 59 SynDD drivers	Done
5	Test on unseen data (3MDAD, DAD)	Done
6	Guided recording tool (prompts + saved times)	Done
7	Consent form	Planned
8	Record 10–20 volunteers, day and night	Planned
9	Run all scenarios on the recordings	Planned
10	Measure speed on the Raspberry Pi with all models	Planned
11	Write the final results report	Planned

10. References

[1] Google. MediaPipe Face Mesh V2 and BlazeFace model cards; MediaPipe Face Landmarker documentation. ai.google.dev/edge/mediapipe

[2] T. Soukupová, J. Čech. "Real-Time Eye Blink Detection using Facial Landmarks." CVWW, 2016.

[3] G. Jocher, J. Qiu. Ultralytics YOLO26. docs.ultralytics.com, 2025.

[4] RTDMS project team. Real-Time Driver Monitoring System, Project 1 Final Report. AASTMT, 2026.

[5] F. Song, X. Tan, X. Liu, S. Chen. "Eyes Closeness Detection from Still Images with Multi-scale Histograms of Principal Oriented Gradients." Pattern Recognition, 47(9):2825–2838, 2014. (CEW)

[6] R. Fusek. "Pupil Localization Using Geodesic Distance." ISVC 2018, Springer, 2018. (MRL Eye Dataset)

[7] R. Ghoddoosian, M. Galib, V. Athitsos. "A Realistic Dataset and Baseline Temporal Model for Early Drowsiness Detection." IEEE CVPR Workshops, 2019. (UTA-RLDD)

[8] D. F. Dinges, R. Grace. "PERCLOS: A Valid Psychophysiological Measure of Alertness as Assessed by Psychomotor Vigilance." US FHWA Tech Brief FHWA-MCRT-98-006, 1998.

[9] DG-Unicamp Driver Gaze Dataset, University of Campinas (as selected in the Project 1 report).

[10] M. S. Rahman et al. "Synthetic Distracted Driving (SynDD2) Dataset for Analyzing Distracted Behaviors and Various Gaze Zones of a Driver." arXiv:2204.08096, 2022. (AI City Challenge 2024 Track 3)

[11] I. Jegham, A. Ben Khalifa, I. Alouani, M. A. Mahjoub. "A Novel Public Dataset for Multimodal Multiview and Multispectral Driver Distraction Analysis: 3MDAD." Signal Processing: Image Communication, 88:115960, 2020.

[12] O. Köpüklü, J. Zheng, H. Xu, G. Rigoll. "Driver Anomaly Detection: A Dataset and Contrastive Learning Approach." IEEE WACV, 2021. (DAD)

[13] CVAT.ai Corporation. Computer Vision Annotation Tool (CVAT). github.com/cvat-ai/cvat

[14] P. Viola, M. Jones. "Rapid Object Detection using a Boosted Cascade of Simple Features." IEEE CVPR, 2001.

[15] W. Wu, H. Peng, S. Yu. "YuNet: A Tiny Millisecond-level Face Detector." Machine Intelligence Research, 20(5):656–665, 2023.

[16] State Farm. State Farm Distracted Driver Detection, Kaggle, 2016. (Used in version 1 only.)